Morphic Studio

Document 24 — Workflow Engine & Automation (Expanded Specification)

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Version: 2.0

Status: Expanded Specification

Priority: Core Infrastructure

Purpose

The Workflow Engine is the execution system that manages every automated process inside Morphic Studio.

It coordinates AI models, production tasks, background jobs, asset generation, rendering, exports, retries, checkpoints, and creator approvals.

Rather than treating every AI request as an isolated event, the Workflow Engine executes structured workflows that may contain hundreds of dependent tasks.

Vision

The Workflow Engine should behave like an experienced production manager.

It understands:

- What needs to happen.
- When it should happen.
- What depends on previous work.
- What can happen simultaneously.
- What requires creator approval.
- How to recover from failures.

The creator focuses on storytelling while the platform coordinates production.

Core Philosophy

Every Creative Process Is a Workflow

Nothing happens randomly.

Every production activity is represented as a structured workflow with defined states, dependencies, validations, and checkpoints.

Objectives

The Workflow Engine shall:

- Execute production workflows.
- Coordinate AI tasks.
- Track dependencies.
- Manage queues.
- Support background processing.
- Resume interrupted work.
- Retry failed operations.
- Notify creators of progress.
- Maintain workflow history.

Workflow Lifecycle

Every workflow passes through the following states:

Draft

↓

Validated

↓

Queued

↓

Preparing

↓

Running

↓

Waiting

↓

Creator Review

↓

Approved

↓

Completed

↓

Archived

If an issue occurs:

↓

Paused

↓

Retry

↓

Recovered

or

↓

Failed

Workflow Types

Supported workflow categories include:

Story Workflows

- Story analysis
- Story Bible updates
- Character extraction
- World extraction

Asset Workflows

- Character generation
- Outfit generation
- Background generation
- Prop generation
- Voice generation

Production Workflows

- Storyboarding
- Comic generation
- Animation production
- Motion comic production

Export Workflows

- PDF export
- Image export
- Video rendering
- Webtoon export
- Archive packaging

Maintenance Workflows

- Asset cleanup
- Cache rebuilding
- Index optimization
- Backup creation

Workflow Structure

Each workflow contains:

- Workflow ID
- Name
- Description
- Creator
- Status
- Priority
- Creation Time
- Start Time
- Completion Time
- Dependencies
- Assigned AI Models
- Estimated Duration
- Progress
- Logs

Task Graph

A workflow consists of tasks connected through dependencies.

Example:

Analyze Story

↓

Update Story Bible

↓

Identify Missing Characters

↓

Generate Characters

↓

Generate Outfits

↓

Generate Backgrounds

↓

Create Storyboard

↓

Creator Approval

↓

Generate Comic

↓

Export

Independent tasks may execute simultaneously.

Parallel Execution

The engine should detect opportunities for parallel processing.

Example:

Generate Character A

||

Generate Character B

||

Generate Backgrounds

||

Generate Props

↓

Merge Results

↓

Continue Workflow

This reduces production time.

Checkpoints

Important milestones become checkpoints.

Examples:

- Story approved
- Storyboard approved
- Character approved
- Asset generation complete
- Comic generation complete
- Animation review complete

Creators can resume from any checkpoint.

Automation Rules

Automation may include:

- Auto-save
- Auto-analysis
- Auto-validation
- Auto-retry
- Auto-notification
- Scheduled rendering
- Automatic asset reuse
- Batch processing

Automation rules remain configurable.

User Stories

As a creator, I want long productions to continue in the background.

As a creator, I want failed tasks to resume without restarting everything.

As a creator, I want to pause production and continue later.

As a creator, I want to understand what the system is doing at every stage.

Functional Requirements

The Workflow Engine shall allow users to:

- Start workflows.
- Pause workflows.
- Resume workflows.
- Cancel workflows.
- Duplicate workflows.
- Monitor progress.
- Retry failed tasks.
- View execution logs.
- Configure automation.
- Schedule workflows.

Workflow Dashboard

The dashboard should display:

- Active workflows
- Completed workflows
- Failed workflows
- Pending approvals
- Queue length
- Estimated completion times
- AI resource usage
- System health

Notification System

Creators should receive notifications for:

- Workflow started
- Workflow completed

- Creator approval required
- Errors detected
- Asset ready
- Export complete
- Retry completed

Notifications may appear in-app and through supported channels.

Database Entities

Core entities include:

- Workflow
- WorkflowTask
- TaskDependency
- WorkflowCheckpoint
- WorkflowHistory
- WorkflowQueue
- ExecutionLog
- Notification

API Considerations

Potential endpoints include:

- Create Workflow
- Start Workflow
- Pause Workflow
- Resume Workflow
- Cancel Workflow
- Retrieve Workflow
- Workflow Logs
- Workflow Status
- Queue Status

Non-Functional Requirements

The Workflow Engine should:

- Scale to thousands of tasks.
- Execute reliably.

- Recover automatically.
- Support distributed processing.
- Maintain execution history.
- Minimize downtime.

Error Handling

If a task fails:

- Preserve completed work.
- Retry according to policy.
- Notify the creator if intervention is required.
- Continue independent tasks when possible.

If an AI provider becomes unavailable:

- Switch to an approved fallback model.
- Resume execution automatically.

Future Expansion

Future versions may include:

- Visual workflow editor.
- Drag-and-drop automation builder.
- Creator-defined automation rules.
- Team workflow orchestration.
- AI workflow optimization.
- Predictive scheduling.
- Cross-project automation.
- Marketplace workflow templates.

Acceptance Criteria

The Workflow Engine succeeds when creators can execute complex productions involving hundreds of dependent tasks without needing to manually coordinate AI models, background jobs, or production stages.

Key Principle

Creative work should feel effortless. The Workflow Engine quietly manages complexity behind the scenes, ensuring every task happens at the right time, in the right order, with the right resources, while creators remain focused on storytelling.

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